Project Review 2: Database Design & SQL Implementation

Course & Project Overview

- **Course:** DATABASE SYSTEMS
- **Project Title:** Student University Management System (UMS)
- **Team Members:**
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1. ER Diagram to Relational Table Conversion

Primary Keys and Foreign Keys

- **DEPARTMENT**
    - **Primary Key:** dept_id
- **PROGRAMME**
    - **Primary Key:** programme_id
    - **Foreign Key:** dept_id references DEPARTMENT(dept_id)
- **FACULTY**
    - **Primary Key:** faculty_id
    - **Foreign Key:** dept_id references DEPARTMENT(dept_id)
- **SEMESTER**
    - **Primary Key:** semester_id
- **COURSE**
    - **Primary Key:** course_id
    - **Foreign Keys:** dept_id references DEPARTMENT(dept_id), prereq_course_id references COURSE(course_id)

- **HOSTEL_BLOCK**
  - **Primary Key:** block_id
- **HOSTEL_ROOM**
  - **Primary Key:** room_id
  - **Foreign Key:** block_id references HOSTEL_BLOCK(block_id)
- **STUDENT**
  - **Primary Key:** student_id
  - **Foreign Keys:** programme_id references PROGRAMME(programme_id), advisor_id references FACULTY(faculty_id), room_id references HOSTEL_ROOM(room_id)
- **COURSE_WAITLIST**
  - **Primary Key:** waitlist_id
  - **Foreign Keys:** student_id references STUDENT(student_id), course_id references COURSE(course_id)
- **FEE_PAYMENT**
  - **Primary Key:** payment_id
  - **Foreign Key:** student_id references STUDENT(student_id)
- **ENROLLMENT** (Weak Entity)
  - **Primary Key:** Composite (student_id, course_id)
  - **Foreign Keys:** student_id references STUDENT(student_id), course_id references COURSE(course_id), semester_id references SEMESTER(semester_id)
- **ATTENDANCE** (Associative Entity)
  - **Primary Key:** Composite (student_id, course_id, attendance_date)
  - **Foreign Keys:** student_id references STUDENT(student_id), course_id references COURSE(course_id)

2. Data Integrity and Constraints

Table	Attribute	Constraints	Reason
DEPARTMENT	dept_id	PRIMARY KEY, NOT NULL	Uniquely identifies each academic department

Table	Attribute	Constraints	Reason
DEPARTMENT	dept_code	UNIQUE, NOT NULL	Prevents duplicate department abbreviations/codes
DEPARTMENT	dept_name	NOT NULL	Ensures every department has an official recorded title
PROGRAMME	programme_id	PRIMARY KEY, NOT NULL	Uniquely identifies each academic degree program
PROGRAMME	programme_code	UNIQUE, NOT NULL	Unique institutional identifier for curriculum offerings
PROGRAMME	dept_id	FOREIGN KEY, NOT NULL	Enforces department ownership; restricted on delete
FACULTY	faculty_id	PRIMARY KEY, NOT NULL	Uniquely identifies each faculty instructor
FACULTY	employee_code	UNIQUE, NOT NULL	Enforces uniqueness for payroll/faculty ID credentials
FACULTY	email	UNIQUE, NOT NULL	Prevents duplicate institutional email addresses
FACULTY	dept_id	FOREIGN KEY, NOT NULL	References employing department; restricted on delete
SEMESTER	semester_id	PRIMARY KEY, NOT NULL	Uniquely identifies each academic term
SEMESTER	academic_year, term	NOT NULL	Mandates term classification for chronological scheduling

Table	Attribute	Constraints	Reason
COURSE	course_id	PRIMARY KEY, NOT NULL	Uniquely identifies a specific course section offering
COURSE	course_code	UNIQUE, NOT NULL	Guarantees catalog course code uniqueness
COURSE	dept_id	FOREIGN KEY, NOT NULL	Links course offering to parent academic department
COURSE	prereq_course_id	FOREIGN KEY	Self-referencing link; set null on course removal
HOSTEL_BLOCK	block_id	PRIMARY KEY, NOT NULL	Uniquely identifies each residential block
HOSTEL_BLOCK	block_name	UNIQUE, NOT NULL	Prevents duplicate naming of campus hostel facilities
HOSTEL_ROOM	room_id	PRIMARY KEY, NOT NULL	Uniquely identifies each physical dormitory room
HOSTEL_ROOM	block_id	FOREIGN KEY, NOT NULL	Links room to its physical block; cascades on deletion
HOSTEL_ROOM	(block_id, room_number)	UNIQUE	Disallows duplicate room identifiers within the same hostel block
STUDENT	student_id	PRIMARY KEY, NOT NULL	Uniquely identifies individual student profile records
STUDENT	reg_no	UNIQUE, NOT NULL	Enforces uniqueness of student registration number

Table	Attribute	Constraints	Reason
STUDENT	email	UNIQUE, NOT NULL	Institutional communication email must be unique per student
STUDENT	programme_id	FOREIGN KEY, NOT NULL	Enforces mandatory enrollment in an active degree program
STUDENT	advisor_id	FOREIGN KEY	Maps faculty proctor/advisor; set to null if faculty departs
STUDENT	room_id	FOREIGN KEY	Links hostel room allotment; remains null for day scholars
ENROLLMENT	(student_id, course_id)	PRIMARY KEY, FOREIGN...	Composite key identifying registration; cascades on student/course removal
ENROLLMENT	semester_id	FOREIGN KEY, NOT NULL	Links registration directly to active academic term
ATTENDANCE	(student_id, course_id, a...	PRIMARY KEY, FOREIGN...	Composite key uniquely logging session presence per day
COURSE_WAITLIST	waitlist_id	PRIMARY KEY, NOT NULL	Uniquely identifies queue entry for automated seat allocation
COURSE_WAITLIST	(student_id, course_id)	UNIQUE, FOREIGN KEY	Disallows duplicate waitlist registrations by the same student
FEE_PAYMENT	payment_id	PRIMARY KEY, NOT NULL	Uniquely identifies student tuition/hostel payment records

Table	Attribute	Constraints	Reason
FEE_PAYMENT	invoice_no	UNIQUE, NOT NULL	Audit and financial tracking invoice reference key
FEE_PAYMENT	student_id	FOREIGN KEY, NOT NULL	Binds billing transaction to student account; cascades on deletion

3. Queries for creation (DDL - tables and views)

Table Creation

```
CREATE TABLE DEPARTMENT (
    dept_id INT AUTO_INCREMENT,
    dept_code VARCHAR(10) NOT NULL,
    dept_name VARCHAR(100) NOT NULL,
    building VARCHAR(50) NOT NULL,
    budget DECIMAL(12,2) NOT NULL DEFAULT 0.00,
    CONSTRAINT pk_department PRIMARY KEY (dept_id),
    CONSTRAINT uk_dept_code UNIQUE (dept_code)
);

CREATE TABLE PROGRAMME (
```

```
    programme_id INT AUTO_INCREMENT,
    programme_code VARCHAR(15) NOT NULL,
    programme_name VARCHAR(100) NOT NULL,
    degree_level VARCHAR(20) NOT NULL,
    total_credits_required INT NOT NULL,
    dept_id INT NOT NULL,
    CONSTRAINT pk_programme PRIMARY KEY (programme_id),
    CONSTRAINT uk_programme_code UNIQUE (programme_code),
    CONSTRAINT fk_programme_dept FOREIGN KEY (dept_id)
        REFERENCES DEPARTMENT(dept_id) ON DELETE RESTRICT
);
CREATE TABLE FACULTY (
    faculty_id INT AUTO_INCREMENT,
    employee_code VARCHAR(15) NOT NULL,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    email VARCHAR(100) NOT NULL,
    phone VARCHAR(15) NOT NULL,
```

```
    designation VARCHAR(50) NOT NULL,
    dept_id INT NOT NULL,
    CONSTRAINT pk_faculty PRIMARY KEY (faculty_id),
    CONSTRAINT uk_faculty_emp_code UNIQUE (employee_code),
    CONSTRAINT uk_faculty_email UNIQUE (email),
    CONSTRAINT fk_faculty_dept FOREIGN KEY (dept_id)
        REFERENCES DEPARTMENT(dept_id) ON DELETE RESTRICT
);
CREATE TABLE SEMESTER (
    semester_id INT AUTO_INCREMENT,
    academic_year VARCHAR(10) NOT NULL,
    term VARCHAR(10) NOT NULL,
    start_date DATE NOT NULL,
    end_date DATE NOT NULL,
    is_active TINYINT(1) NOT NULL DEFAULT 0,
    CONSTRAINT pk_semester PRIMARY KEY (semester_id)
);
CREATE TABLE COURSE (
```

```
    course_id INT AUTO_INCREMENT,
    course_code VARCHAR(15) NOT NULL,
    course_title VARCHAR(100) NOT NULL,
    credits INT NOT NULL,
    capacity INT NOT NULL,
    available_seats INT NOT NULL,
    dept_id INT NOT NULL,
    prereq_course_id INT NULL,
    CONSTRAINT pk_course PRIMARY KEY (course_id),
    CONSTRAINT uk_course_code UNIQUE (course_code),
    CONSTRAINT fk_course_dept FOREIGN KEY (dept_id)
        REFERENCES DEPARTMENT(dept_id) ON DELETE RESTRICT,
    CONSTRAINT fk_course_prereq FOREIGN KEY (prereq_course_id)
        REFERENCES COURSE(course_id) ON DELETE SET NULL
);
CREATE TABLE HOSTEL_BLOCK (
    block_id INT AUTO_INCREMENT,
    block_name VARCHAR(50) NOT NULL,
```

```
    gender_type CHAR(1) NOT NULL,
    total_capacity INT NOT NULL,
    CONSTRAINT pk_hostel_block PRIMARY KEY (block_id),
    CONSTRAINT uk_block_name UNIQUE (block_name)
);
CREATE TABLE HOSTEL_ROOM (
    room_id INT AUTO_INCREMENT,
    block_id INT NOT NULL,
    room_number VARCHAR(10) NOT NULL,
    floor_number INT NOT NULL,
    capacity INT NOT NULL DEFAULT 2,
    occupied_beds INT NOT NULL DEFAULT 0,
    monthly_rent DECIMAL(8,2) NOT NULL,
    CONSTRAINT pk_hostel_room PRIMARY KEY (room_id),
    CONSTRAINT uk_block_room UNIQUE (block_id, room_number),
    CONSTRAINT fk_room_block FOREIGN KEY (block_id)
        REFERENCES HOSTEL_BLOCK(block_id) ON DELETE CASCADE
);
```

```
CREATE TABLE STUDENT (
    student_id INT AUTO_INCREMENT,
    reg_no VARCHAR(15) NOT NULL,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    email VARCHAR(100) NOT NULL,
    dob DATE NOT NULL,
    gender CHAR(1) NOT NULL,
    cgpa DECIMAL(3,2) NOT NULL DEFAULT 0.00,
    programme_id INT NOT NULL,
    advisor_id INT NULL,
    room_id INT NULL,
    CONSTRAINT pk_student PRIMARY KEY (student_id),
    CONSTRAINT uk_student_reg_no UNIQUE (reg_no),
    CONSTRAINT uk_student_email UNIQUE (email),
    CONSTRAINT fk_student_programme FOREIGN KEY (programme_id)
        REFERENCES PROGRAMME(programme_id) ON DELETE RESTRICT,
    CONSTRAINT fk_student_advisor FOREIGN KEY (advisor_id)
```

```
        REFERENCES FACULTY(faculty_id) ON DELETE SET NULL,
    CONSTRAINT fk_student_room FOREIGN KEY (room_id)
        REFERENCES HOSTEL_ROOM(room_id) ON DELETE SET NULL
);
CREATE TABLE ENROLLMENT (
    student_id INT NOT NULL,
    course_id INT NOT NULL,
    semester_id INT NOT NULL,
    enrollment_date DATE NOT NULL,
    grade_points DECIMAL(3,2) NULL,
    completion_status VARCHAR(20) NOT NULL DEFAULT 'ENROLLED',
    CONSTRAINT pk_enrollment PRIMARY KEY (student_id, course_id),
    CONSTRAINT fk_enrollment_student FOREIGN KEY (student_id)
        REFERENCES STUDENT(student_id) ON DELETE CASCADE,
    CONSTRAINT fk_enrollment_course FOREIGN KEY (course_id)
        REFERENCES COURSE(course_id) ON DELETE CASCADE,
    CONSTRAINT fk_enrollment_semester FOREIGN KEY (semester_id)
        REFERENCES SEMESTER(semester_id) ON DELETE RESTRICT
```

```
);
CREATE TABLE ATTENDANCE (
    student_id INT NOT NULL,
    course_id INT NOT NULL,
    attendance_date DATE NOT NULL,
    status VARCHAR(10) NOT NULL,
    CONSTRAINT pk_attendance PRIMARY KEY (student_id, course_id, attendance_date),
    CONSTRAINT fk_attendance_student FOREIGN KEY (student_id)
        REFERENCES STUDENT(student_id) ON DELETE CASCADE,
    CONSTRAINT fk_attendance_course FOREIGN KEY (course_id)
        REFERENCES COURSE(course_id) ON DELETE CASCADE
);
CREATE TABLE COURSE_WAITLIST (
    waitlist_id INT AUTO_INCREMENT,
    student_id INT NOT NULL,
    course_id INT NOT NULL,
    waitlist_time TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP(3),
    priority_position INT NOT NULL,
```

```
    status VARCHAR(15) NOT NULL DEFAULT 'ACTIVE',
    CONSTRAINT pk_course_waitlist PRIMARY KEY (waitlist_id),
    CONSTRAINT uk_student_course_waitlist UNIQUE (student_id, course_id),
    CONSTRAINT fk_waitlist_student FOREIGN KEY (student_id)
        REFERENCES STUDENT(student_id) ON DELETE CASCADE,
    CONSTRAINT fk_waitlist_course FOREIGN KEY (course_id)
        REFERENCES COURSE(course_id) ON DELETE CASCADE
);
CREATE TABLE FEE_PAYMENT (
    payment_id INT AUTO_INCREMENT,
    student_id INT NOT NULL,
    invoice_no VARCHAR(20) NOT NULL,
    total_amount DECIMAL(10,2) NOT NULL,
    paid_amount DECIMAL(10,2) NOT NULL DEFAULT 0.00,
    due_date DATE NOT NULL,
    payment_status VARCHAR(15) NOT NULL DEFAULT 'PENDING',
    payment_date DATE NULL,
    CONSTRAINT pk_fee_payment PRIMARY KEY (payment_id),
```

```
    CONSTRAINT uk_invoice_no UNIQUE (invoice_no),
    CONSTRAINT fk_fee_student FOREIGN KEY (student_id)
        REFERENCES STUDENT(student_id) ON DELETE CASCADE
);
```

```
mysql> SELECT * FROM DEPARTMENT;
+---------+-----------+-----------------------------------------+-----------------------+-------------+
| dept_id | dept_code | dept_name                               | building              | budget      |
+---------+-----------+-----------------------------------------+-----------------------+-------------+
|       1 | SCOPE     | School of Computer Science and Engineering | Technology Tower      | 15000000.00 |
|       2 | SENSE     | School of Electronics Engineering          | M. Visvesvaraya Block | 12000000.00 |
|       3 | SMEC      | School of Mechanical Engineering           | GD Naidu Block        | 10000000.00 |
+---------+-----------+-----------------------------------------+-----------------------+-------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM PROGRAMME;
+--------------+----------------+-----------------------------------------------------+--------------+-------------------------+---------+
| programme_id | programme_code | programme_name                                      | degree_level | total_credits_required | dept_id |
+--------------+----------------+-----------------------------------------------------+--------------+-------------------------+---------+
|            1 | BTECH-CSE-AIML | B.Tech in Computer Science and Engineering (AI & ML) | Undergraduate |                     160 |       1 |
|            2 | BTECH-CSE-CORE | B.Tech in Computer Science and Engineering          | Undergraduate |                     160 |       1 |
|            3 | BTECH-ECE      | B.Tech in Electronics and Communication Engineering  | Undergraduate |                     160 |       2 |
+--------------+----------------+-----------------------------------------------------+--------------+-------------------------+---------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM FACULTY;
+------------+---------------+------------+-----------+--------------------------+------------+---------------------+---------+
| faculty_id | employee_code | first_name | last_name | email                    | phone      | designation         | dept_id |
+------------+---------------+------------+-----------+--------------------------+------------+---------------------+---------+
|          1 | EMP1001       | Ramesh     | Kumar     | ramesh.kumar@vit.ac.in   | 9876543210 | Professor           |       1 |
|          2 | EMP1002       | Priya      | Sundaram  | priya.sundaram@vit.ac.in | 9876543211 | Associate Professor |       1 |
|          3 | EMP1003       | Anand      | Venkatesh | anand.venkatesh@vit.ac.in | 9876543212 | Assistant Professor |       2 |
+------------+---------------+------------+-----------+--------------------------+------------+---------------------+---------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM SEMESTER;
+-------------+---------------+--------+------------+------------+-----------+
| semester_id | academic_year | term   | start_date | end_date   | is_active |
+-------------+---------------+--------+------------+------------+-----------+
|           1 | 2026-2027     | Fall   | 2026-07-15 | 2026-11-30 |         1 |
|           2 | 2026-2027     | Winter | 2026-12-15 | 2027-04-30 |         0 |
+-------------+---------------+--------+------------+------------+-----------+
2 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM COURSE;
+-----------+-------------+-----------------------------------------+---------+----------+-----------------+---------+-----------------+
| course_id | course_code | course_title                            | credits | capacity | available_seats | dept_id | prereq_course_id |
+-----------+-------------+-----------------------------------------+---------+----------+-----------------+---------+-----------------+
|         1 | CSE1001     | Problem Solving and Object-Oriented Programming |       4 |       60 |               0 |       1 |            NULL |
|         2 | CSE2001     | Data Structures and Algorithms          |       4 |       60 |               0 |       1 |               1 |
|         3 | CSE2005     | Database Management Systems             |       4 |       60 |               2 |       1 |               2 |
|         4 | ECE1002     | Semiconductor Devices and Circuits      |       3 |       60 |              15 |       2 |            NULL |
+-----------+-------------+-----------------------------------------+---------+----------+-----------------+---------+-----------------+
4 rows in set (0.00 sec)

mysql> SELECT * FROM HOSTEL_BLOCK;
+----------+---------------+-------------+----------------+
| block_id | block_name    | gender_type | total_capacity |
+----------+---------------+-------------+----------------+
|        1 | Men Block A   | M           |            500 |
|        2 | Ladies Block B | F           |            400 |
+----------+---------------+-------------+----------------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM HOSTEL_ROOM;
+---------+----------+-------------+--------------+----------+---------------+--------------+
| room_id | block_id | room_number | floor_number | capacity | occupied_beds | monthly_rent |
+---------+----------+-------------+--------------+----------+---------------+--------------+
|       1 |        1 | A-201       |            2 |        2 |             2 |      8500.00 |
|       2 |        1 | A-202       |            2 |        2 |             1 |      8500.00 |
|       3 |        2 | B-105       |            1 |        2 |             1 |      9500.00 |
+---------+----------+-------------+--------------+----------+---------------+--------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM STUDENT;
+------------+-----------+------------+-----------+-----------------------------------+------------+--------+------+--------------+------------+---------+
| student_id | reg_no    | first_name | last_name | email                             | dob        | gender | cgpa | programme_id | advisor_id | room_id |
+------------+-----------+------------+-----------+-----------------------------------+------------+--------+------+--------------+------------+---------+
|          1 | 25BAI0077 | Jacob      | Jean      | jacob.jean2025@vitstudent.ac.in   | 2005-04-12 | M      | 9.45 |            1 |          1 |       1 |
|          2 | 25BCE0634 | Aradhya    | Jain      | aradhya.jain2025@vitstudent.ac.in | 2005-08-22 | F      | 9.12 |            2 |          2 |       3 |
|          3 | 25BAI0030 | Krrish     | Pare      | krrish.pare2025@vitstudent.ac.in  | 2005-01-18 | M      | 8.85 |            1 |          1 |       1 |
|          4 | 25ECE0122 | Nikhil     | Sharma    | nikhil.sharma2025@vitstudent.ac.in | 2004-11-05 | M      | 7.90 |            3 |          3 |       2 |
|          5 | 25BCE0899 | Sneha      | Rao       | sneha.rao2025@vitstudent.ac.in    | 2005-03-30 | F      | 8.20 |            2 |          2 |    NULL |
+------------+-----------+------------+-----------+-----------------------------------+------------+--------+------+--------------+------------+---------+
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM ENROLLMENT;
+------------+-----------+-------------+-----------------+--------------+-------------------+
| student_id | course_id | semester_id | enrollment_date | grade_points | completion_status |
+------------+-----------+-------------+-----------------+--------------+-------------------+
|          1 |         1 |           1 | 2026-07-20      |        10.00 | COMPLETED         |
|          1 |         2 |           1 | 2026-07-20      |         9.50 | COMPLETED         |
|          1 |         3 |           1 | 2026-07-21      |         NULL | ENROLLED          |
|          2 |         1 |           1 | 2026-07-20      |         9.00 | COMPLETED         |
|          2 |         3 |           1 | 2026-07-21      |         NULL | ENROLLED          |
|          3 |         3 |           1 | 2026-07-21      |         NULL | ENROLLED          |
|          4 |         4 |           1 | 2026-07-22      |         NULL | ENROLLED          |
+------------+-----------+-------------+-----------------+--------------+-------------------+
7 rows in set (0.00 sec)

mysql> SELECT * FROM ATTENDANCE;
+------------+-----------+-----------------+---------+
| student_id | course_id | attendance_date | status  |
+------------+-----------+-----------------+---------+
|          1 |         3 | 2026-08-01      | Present |
|          1 |         3 | 2026-08-03      | Present |
|          1 |         3 | 2026-08-05      | Present |
|          2 |         3 | 2026-08-01      | Present |
|          2 |         3 | 2026-08-03      | Absent  |
|          2 |         3 | 2026-08-05      | Present |
|          3 |         3 | 2026-08-01      | Absent  |
|          3 |         3 | 2026-08-03      | Absent  |
|          3 |         3 | 2026-08-05      | Present |
+------------+-----------+-----------------+---------+
9 rows in set (0.00 sec)

mysql> SELECT * FROM COURSE_WAITLIST;
+-------------+------------+-----------+-------------------------+-------------------+--------+
| waitlist_id | student_id | course_id | waitlist_time           | priority_position | status |
+-------------+------------+-----------+-------------------------+-------------------+--------+
|           1 |          4 |         2 | 2026-07-21 09:30:15.120 |                 1 | ACTIVE |
|           2 |          5 |         2 | 2026-07-21 09:45:22.450 |                 2 | ACTIVE |
+-------------+------------+-----------+-------------------------+-------------------+--------+
2 rows in set (0.00 sec)

mysql>
```

View Creation

CREATE VIEW v_admin_student_directory AS

SELECT

s.student_id,

s.reg_no,

CONCAT(s.first_name, ' ', s.last_name) AS student_name,

s.email,

p.programme_name,

d.dept_name,

```
    CONCAT(f.first_name, ' ', f.last_name) AS faculty_advisor,
    s.cgpa
FROM STUDENT s
JOIN PROGRAMME p ON s.programme_id = p.programme_id
JOIN DEPARTMENT d ON p.dept_id = d.dept_id
LEFT JOIN FACULTY f ON s.advisor_id = f.faculty_id;
CREATE VIEW v_admin_course_waitlist AS
SELECT
    c.course_id,
    c.course_code,
    c.course_title,
    d.dept_code,
    c.capacity,
    c.available_seats,
    (c.capacity - c.available_seats) AS enrolled_students,
    COUNT(w.waitlist_id) AS total_waitlisted
FROM COURSE c
JOIN DEPARTMENT d ON c.dept_id = d.dept_id
```

```
LEFT JOIN COURSE_WAITLIST w ON c.course_id = w.course_id AND w.status = 'ACTIVE'

GROUP BY c.course_id, c.course_code, c.course_title, d.dept_code, c.capacity,
c.available_seats;

CREATE VIEW v_faculty_view AS

SELECT

    c.course_code,

    c.course_title,

    s.reg_no,

    CONCAT(s.first_name, ' ', s.last_name) AS student_name,

    s.email AS student_email,

    e.enrollment_date,

    e.completion_status

FROM ENROLLMENT e

JOIN COURSE c ON e.course_id = c.course_id

JOIN STUDENT s ON e.student_id = s.student_id

WHERE e.completion_status = 'ENROLLED';

CREATE VIEW v_faculty_attendance_shortage AS

SELECT
```

```
    c.course_code,
    c.course_title,
    s.reg_no,
    CONCAT(s.first_name, ' ', s.last_name) AS student_name,
    s.email AS student_email,
    COUNT(a.attendance_date) AS total_classes_conducted,
    SUM(CASE WHEN a.status = 'Present' THEN 1 ELSE 0 END) AS classes_attended,
    ROUND((SUM(CASE WHEN a.status = 'Present' THEN 1 ELSE 0 END) /
COUNT(a.attendance_date)) * 100, 2) AS attendance_percentage
FROM ATTENDANCE a
JOIN COURSE c ON a.course_id = c.course_id
JOIN STUDENT s ON a.student_id = s.student_id
GROUP BY c.course_id, c.course_code, c.course_title, s.student_id, s.reg_no,
s.first_name, s.last_name
HAVING attendance_percentage < 75.00;
CREATE VIEW v_accounts_fee_summary AS
SELECT
    f.payment_status,
    COUNT(f.payment_id) AS total_invoices,
```

```
    SUM(f.total_amount) AS total_billed_amount,
    SUM(f.paid_amount) AS total_collected_amount,
    SUM(f.total_amount - f.paid_amount) AS outstanding_balance
FROM FEE_PAYMENT f
GROUP BY f.payment_status;
CREATE VIEW v_accounts_defaulters_list AS
SELECT
    s.reg_no,
    CONCAT(s.first_name, ' ', s.last_name) AS student_name,
    s.email,
    f.invoice_no,
    f.total_amount,
    f.paid_amount,
    (f.total_amount - f.paid_amount) AS balance_due,
    f.due_date,
    f.payment_status
FROM FEE_PAYMENT f
JOIN STUDENT s ON f.student_id = s.student_id
```

```
WHERE f.payment_status IN ('PENDING', 'PARTIAL', 'OVERDUE');

CREATE VIEW v_warden_occupancy_overview AS

SELECT

    hb.block_name,

    hb.gender_type,

    hr.room_number,

    hr.floor_number,

    hr.capacity,

    hr.occupied_beds,

    (hr.capacity - hr.occupied_beds) AS vacant_beds,

    hr.monthly_rent

FROM HOSTEL_ROOM hr

JOIN HOSTEL_BLOCK hb ON hr.block_id = hb.block_id

ORDER BY hb.block_name, hr.room_number;

CREATE VIEW v_warden_resident_directory AS

SELECT

    hb.block_name,

    hr.room_number,
```

```
    s.reg_no,
    CONCAT(s.first_name, ' ', s.last_name) AS student_name,
    s.gender,
    s.email,
    p.programme_name
FROM STUDENT s
JOIN HOSTEL_ROOM hr ON s.room_id = hr.room_id
JOIN HOSTEL_BLOCK hb ON hr.block_id = hb.block_id
JOIN PROGRAMME p ON s.programme_id = p.programme_id
ORDER BY hb.block_name, hr.room_number, s.reg_no;
```

```
mysql> SELECT * FROM v_admin_student_directory;
+------------+-----------+---------------+-------------------------------+-----------------------------------------------------+----------------------
----------------------+-----------------+------+
| student_id | reg_no    | student_name  | email                         | programme_name                                      | dept_name
                     | faculty_advisor | cgpa |
+------------+-----------+---------------+-------------------------------+-----------------------------------------------------+----------------------
----------------------+-----------------+------+
|          1 | 25BAI0077 | Jacob Jean    | jacob.jean2025@vitstudent.ac.in    | B.Tech in Computer Science and Engineering (AI & ML) | School of Computer Sc
ience and Engineering | Ramesh Kumar    | 9.45 |
|          3 | 25BAI0030 | Krrish Pare   | krrish.pare2025@vitstudent.ac.in   | B.Tech in Computer Science and Engineering (AI & ML) | School of Computer Sc
ience and Engineering | Ramesh Kumar    | 8.85 |
|          2 | 25BCE0634 | Aradhya Jain  | aradhya.jain2025@vitstudent.ac.in  | B.Tech in Computer Science and Engineering           | School of Computer Sc
ience and Engineering | Priya Sundaram  | 9.12 |
|          5 | 25BCE0899 | Sneha Rao     | sneha.rao2025@vitstudent.ac.in     | B.Tech in Computer Science and Engineering           | School of Computer Sc
ience and Engineering | Priya Sundaram  | 8.20 |
|          4 | 25ECE0122 | Nikhil Sharma | nikhil.sharma2025@vitstudent.ac.in | B.Tech in Electronics and Communication Engineering  | School of Electronics
 Engineering          | Anand Venkatesh | 7.90 |
+------------+-----------+---------------+-------------------------------+-----------------------------------------------------+----------------------
----------------------+-----------------+------+
5 rows in set (0.06 sec)

mysql> SELECT * FROM v_admin_course_waitlist;
+-----------+-------------+-----------------------------------------------+-----------+----------+-----------------+--------------------+------------------
+
| course_id | course_code | course_title                                  | dept_code | capacity | available_seats | enrolled_students | total_waitlisted
|
+-----------+-------------+-----------------------------------------------+-----------+----------+-----------------+--------------------+------------------
+
|         1 | CSE1001     | Problem Solving and Object-Oriented Programming | SCOPE     |       60 |               0 |                 60 |                0
|
|         2 | CSE2001     | Data Structures and Algorithms                | SCOPE     |       60 |               0 |                 60 |                2
|
|         3 | CSE2005     | Database Management Systems                   | SCOPE     |       60 |               2 |                 58 |                0
|
|         4 | ECE1002     | Semiconductor Devices and Circuits            | SENSE     |       60 |              15 |                 45 |                0
|
+-----------+-------------+-----------------------------------------------+-----------+----------+-----------------+--------------------+------------------
+
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM v_faculty_view;
+-------------+-----------------------------------+-----------+---------------+-------------------------------+-----------------+-------------------+
| course_code | course_title                      | reg_no    | student_name  | student_email                 | enrollment_date | completion_status |
+-------------+-----------------------------------+-----------+---------------+-------------------------------+-----------------+-------------------+
| CSE2005     | Database Management Systems       | 25BAI0077 | Jacob Jean    | jacob.jean2025@vitstudent.ac.in    | 2026-07-21      | ENROLLED          |
| CSE2005     | Database Management Systems       | 25BCE0634 | Aradhya Jain  | aradhya.jain2025@vitstudent.ac.in  | 2026-07-21      | ENROLLED          |
| CSE2005     | Database Management Systems       | 25BAI0030 | Krrish Pare   | krrish.pare2025@vitstudent.ac.in   | 2026-07-21      | ENROLLED          |
| ECE1002     | Semiconductor Devices and Circuits | 25ECE0122 | Nikhil Sharma | nikhil.sharma2025@vitstudent.ac.in | 2026-07-22      | ENROLLED          |
+-------------+-----------------------------------+-----------+---------------+-------------------------------+-----------------+-------------------+
4 rows in set (0.00 sec)

mysql> SELECT * FROM v_faculty_attendance_shortage;
+-------------+-----------------------------+-----------+--------------+-------------------------------+-----------------------+------------------+---
--------------------+
| course_code | course_title                | reg_no    | student_name | student_email                 | total_classes_conducted | classes_attended | at
tendance_percentage |
+-------------+-----------------------------+-----------+--------------+-------------------------------+-----------------------+------------------+---
--------------------+
| CSE2005     | Database Management Systems | 25BCE0634 | Aradhya Jain | aradhya.jain2025@vitstudent.ac.in |                     3 |                2 |
              66.67 |
| CSE2005     | Database Management Systems | 25BAI0030 | Krrish Pare  | krrish.pare2025@vitstudent.ac.in  |                     3 |                1 |
              33.33 |
+-------------+-----------------------------+-----------+--------------+-------------------------------+-----------------------+------------------+---
--------------------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM v_accounts_fee_summary;
+----------------+----------------+--------------------+-----------------------+---------------------+
| payment_status | total_invoices | total_billed_amount | total_collected_amount | outstanding_balance |
+----------------+----------------+--------------------+-----------------------+---------------------+
| PAID           |              2 |          400000.00 |             400000.00 |                0.00 |
| PARTIAL        |              1 |          195000.00 |             100000.00 |            95000.00 |
| OVERDUE        |              1 |          185000.00 |                  0.00 |           185000.00 |
+----------------+----------------+--------------------+-----------------------+---------------------+
3 rows in set (0.01 sec)
```

```
mysql> SELECT * FROM v_accounts_defaulters_list;
+-----------+---------------+-------------------------------+------------+--------------+-------------+-------------+------------+----------------+
| reg_no    | student_name  | email                         | invoice_no | total_amount | paid_amount | balance_due | due_date   | payment_status |
+-----------+---------------+-------------------------------+------------+--------------+-------------+-------------+------------+----------------+
| 25BAI0030 | Krrish Pare   | krrish.pare2025@vitstudent.ac.in   | INV-2026-003 |    195000.00 |   100000.00 |    95000.00 | 2026-08-10 | PARTIAL        |
| 25ECE0122 | Nikhil Sharma | nikhil.sharma2025@vitstudent.ac.in | INV-2026-004 |    185000.00 |        0.00 |   185000.00 | 2026-08-10 | OVERDUE        |
+-----------+---------------+-------------------------------+------------+--------------+-------------+-------------+------------+----------------+
2 rows in set (0.00 sec)

mysql> SELECT * FROM v_warden_occupancy_overview;
+---------------+-------------+-------------+--------------+----------+---------------+-------------+--------------+
| block_name    | gender_type | room_number | floor_number | capacity | occupied_beds | vacant_beds | monthly_rent |
+---------------+-------------+-------------+--------------+----------+---------------+-------------+--------------+
| Ladies Block B | F           | B-105       |            1 |        2 |             1 |           1 |      9500.00 |
| Men Block A    | M           | A-201       |            2 |        2 |             2 |           0 |      8500.00 |
| Men Block A    | M           | A-202       |            2 |        2 |             1 |           1 |      8500.00 |
+---------------+-------------+-------------+--------------+----------+---------------+-------------+--------------+
3 rows in set (0.00 sec)

mysql>
```

4. Sample Insertion (DML)

INSERT INTO DEPARTMENT (dept_code, dept_name, building, budget) VALUES

('SCOPE', 'School of Computer Science and Engineering', 'Technology Tower', 15000000.00),

('SENSE', 'School of Electronics Engineering', 'M. Visvesvaraya Block', 12000000.00),

('SMEC', 'School of Mechanical Engineering', 'GD Naidu Block', 10000000.00);

INSERT INTO PROGRAMME (programme_code, programme_name, degree_level, total_credits_required, dept_id) VALUES

('BTECH-CSE-AIML', 'B.Tech in Computer Science and Engineering (AI & ML)', 'Undergraduate', 160, 1),

('BTECH-CSE-CORE', 'B.Tech in Computer Science and Engineering', 'Undergraduate', 160, 1),

('BTECH-ECE', 'B.Tech in Electronics and Communication Engineering', 'Undergraduate', 160, 2);

INSERT INTO FACULTY (employee_code, first_name, last_name, email, phone, designation, dept_id) VALUES

('EMP1001', 'Ramesh', 'Kumar', 'ramesh.kumar@vit.ac.in', '9876543210', 'Professor', 1),

('EMP1002', 'Priya', 'Sundaram', 'priya.sundaram@vit.ac.in', '9876543211', 'Associate Professor', 1),

('EMP1003', 'Anand', 'Venkatesh', 'anand.venkatesh@vit.ac.in', '9876543212', 'Assistant Professor', 2);

INSERT INTO SEMESTER (academic_year, term, start_date, end_date, is_active) VALUES

('2026-2027', 'Fall', '2026-07-15', '2026-11-30', 1),

('2026-2027', 'Winter', '2026-12-15', '2027-04-30', 0);

INSERT INTO COURSE (course_code, course_title, credits, capacity, available_seats, dept_id, prereq_course_id) VALUES

('CSE1001', 'Problem Solving and Object-Oriented Programming', 4, 60, 0, 1, NULL),

('CSE2001', 'Data Structures and Algorithms', 4, 60, 0, 1, 1),

('CSE2005', 'Database Management Systems', 4, 60, 2, 1, 2),

('ECE1002', 'Semiconductor Devices and Circuits', 3, 60, 15, 2, NULL);

INSERT INTO HOSTEL_BLOCK (block_name, gender_type, total_capacity) VALUES

('Men Block A', 'M', 500),

('Ladies Block B', 'F', 400);

INSERT INTO HOSTEL_ROOM (block_id, room_number, floor_number, capacity, occupied_beds, monthly_rent) VALUES

(1, 'A-201', 2, 2, 2, 8500.00),

(1, 'A-202', 2, 2, 1, 8500.00),

(2, 'B-105', 1, 2, 1, 9500.00);

INSERT INTO STUDENT (reg_no, first_name, last_name, email, dob, gender, cgpa, programme_id, advisor_id, room_id) VALUES

('25BAI0077', 'Jacob', 'Jean', 'jacob.jean2025@vitstudent.ac.in', '2005-04-12', 'M', 9.45, 1, 1, 1),

('25BCE0634', 'Aradhya', 'Jain', 'aradhya.jain2025@vitstudent.ac.in', '2005-08-22', 'F', 9.12, 2, 2, 3),

('25BAI0030', 'Krrish', 'Pare', 'krrish.pare2025@vitstudent.ac.in', '2005-01-18', 'M', 8.85, 1, 1, 1),

('25ECE0122', 'Nikhil', 'Sharma', 'nikhil.sharma2025@vitstudent.ac.in', '2004-11-05', 'M', 7.90, 3, 3, 2),

('25BCE0899', 'Sneha', 'Rao', 'sneha.rao2025@vitstudent.ac.in', '2005-03-30', 'F', 8.20, 2, 2, NULL);

INSERT INTO ENROLLMENT (student_id, course_id, semester_id, enrollment_date, grade_points, completion_status) VALUES

(1, 1, 1, '2026-07-20', 10.00, 'COMPLETED'),

(1, 2, 1, '2026-07-20', 9.50, 'COMPLETED'),

(1, 3, 1, '2026-07-21', NULL, 'ENROLLED'),

(2, 1, 1, '2026-07-20', 9.00, 'COMPLETED'),

(2, 3, 1, '2026-07-21', NULL, 'ENROLLED'),

(3, 3, 1, '2026-07-21', NULL, 'ENROLLED'),

(4, 4, 1, '2026-07-22', NULL, 'ENROLLED');

INSERT INTO ATTENDANCE (student_id, course_id, attendance_date, status) VALUES

(1, 3, '2026-08-01', 'Present'),

(1, 3, '2026-08-03', 'Present'),

(1, 3, '2026-08-05', 'Present'),

(2, 3, '2026-08-01', 'Present'),

(2, 3, '2026-08-03', 'Absent'),

(2, 3, '2026-08-05', 'Present'),

(3, 3, '2026-08-01', 'Absent'),

(3, 3, '2026-08-03', 'Absent'),

(3, 3, '2026-08-05', 'Present');

INSERT INTO COURSE_WAITLIST (student_id, course_id, waitlist_time, priority_position, status) VALUES

(4, 2, '2026-07-21 09:30:15.120', 1, 'ACTIVE'),

(5, 2, '2026-07-21 09:45:22.450', 2, 'ACTIVE');

INSERT INTO FEE_PAYMENT (student_id, invoice_no, total_amount, paid_amount, due_date, payment_status, payment_date) VALUES

(1, 'INV-2026-001', 195000.00, 195000.00, '2026-08-10', 'PAID', '2026-08-02'),

(2, 'INV-2026-002', 205000.00, 205000.00, '2026-08-10', 'PAID', '2026-08-05'),

(3, 'INV-2026-003', 195000.00, 100000.00, '2026-08-10', 'PARTIAL', '2026-08-08'),

(4, 'INV-2026-004', 185000.00, 0.00, '2026-08-10', 'OVERDUE', NULL);

5. Sample queries for data selection

Filtering and Sorting

```
SELECT reg_no, first_name, last_name, email, cgpa

FROM STUDENT

WHERE cgpa >= 8.50

ORDER BY cgpa DESC, last_name ASC;
```

Grouping and Aggregation

```
SELECT

    p.programme_code,

    p.programme_name,

    COUNT(s.student_id) AS total_students,

    ROUND(AVG(s.cgpa), 2) AS average_cgpa,

    MAX(s.cgpa) AS highest_cgpa

FROM PROGRAMME p

JOIN STUDENT s ON p.programme_id = s.programme_id

GROUP BY p.programme_id, p.programme_code, p.programme_name

HAVING COUNT(s.student_id) >= 2

ORDER BY average_cgpa DESC;
```

OUTPUT:

```
mysql> SELECT
    ->     p.programme_code,
    ->     p.programme_name,
    ->     COUNT(s.student_id) AS total_students,
    ->     ROUND(AVG(s.cgpa), 2) AS average_cgpa,
    ->     MAX(s.cgpa) AS highest_cgpa
    -> FROM PROGRAMME p
    -> JOIN STUDENT s ON p.programme_id = s.programme_id
    -> GROUP BY p.programme_id, p.programme_code, p.programme_name
    -> HAVING COUNT(s.student_id) >= 2
    -> ORDER BY average_cgpa DESC;
+----------------+-----------------------------------------------------+----------------+--------------+--------------+
| programme_code | programme_name                                      | total_students | average_cgpa | highest_cgpa |
+----------------+-----------------------------------------------------+----------------+--------------+--------------+
| BTECH-CSE-AIML | B.Tech in Computer Science and Engineering (AI & ML) |              2 |         9.15 |         9.45 |
| BTECH-CSE-CORE | B.Tech in Computer Science and Engineering           |              2 |         8.66 |         9.12 |
+----------------+-----------------------------------------------------+----------------+--------------+--------------+
2 rows in set (0.01 sec)
```

Multi-table Joins

SELECT

s.reg_no,

CONCAT(s.first_name, ' ', s.last_name) AS student_name,

c.course_code,

c.course_title,

e.completion_status

FROM ENROLLMENT e

INNER JOIN STUDENT s ON e.student_id = s.student_id

INNER JOIN COURSE c ON e.course_id = c.course_id;

SELECT

s.reg_no,

```
    CONCAT(s.first_name, ' ', s.last_name) AS student_name,
    COALESCE(hb.block_name, 'Non-Resident') AS hostel_block,
    COALESCE(hr.room_number, 'Day Scholar') AS room_allocated
FROM STUDENT s
LEFT JOIN HOSTEL_ROOM hr ON s.room_id = hr.room_id
LEFT JOIN HOSTEL_BLOCK hb ON hr.block_id = hb.block_id
ORDER BY hostel_block ASC, s.reg_no ASC;
```

Output:

```
mysql> SELECT
    ->     s.reg_no,
    ->     CONCAT(s.first_name, ' ', s.last_name) AS student_name,
    ->     COALESCE(hb.block_name, 'Non-Resident') AS hostel_block,
    ->     COALESCE(hr.room_number, 'Day Scholar') AS room_allocated
    -> FROM STUDENT s
    -> LEFT JOIN HOSTEL_ROOM hr ON s.room_id = hr.room_id
    -> LEFT JOIN HOSTEL_BLOCK hb ON hr.block_id = hb.block_id
    -> ORDER BY hostel_block ASC, s.reg_no ASC;
+-----------+--------------+---------------+----------------+
| reg_no    | student_name | hostel_block  | room_allocated |
+-----------+--------------+---------------+----------------+
| 25BCE0634 | Aradhya Jain | Ladies Block B | B-105          |
| 25BAI0030 | Krrish Pare  | Men Block A    | A-201          |
| 25BAI0077 | Jacob Jean   | Men Block A    | A-201          |
| 25ECE0122 | Nikhil Sharma | Men Block A   | A-202          |
| 25BCE0899 | Sneha Rao    | Non-Resident   | Day Scholar    |
+-----------+--------------+---------------+----------------+
5 rows in set (0.00 sec)
```

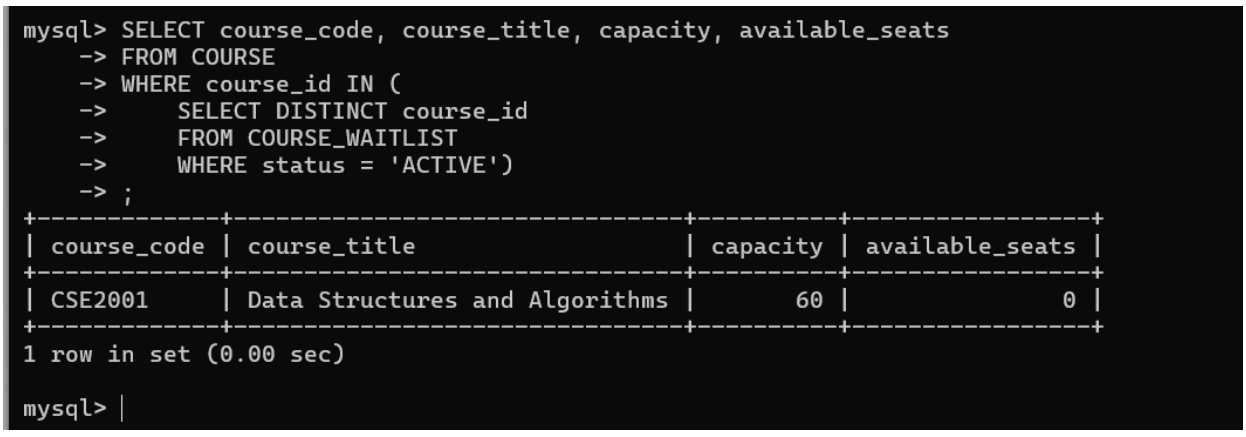
Subqueries

```
SELECT reg_no, first_name, last_name, cgpa
FROM STUDENT
```

```
WHERE cgpa > (
    SELECT AVG(cgpa) FROM STUDENT
)
ORDER BY cgpa DESC;

SELECT course_code, course_title, capacity, available_seats
FROM COURSE
WHERE course_id IN (
    SELECT DISTINCT course_id
    FROM COURSE_WAITLIST
    WHERE status = 'ACTIVE'
);
```

Output:



```
mysql> SELECT course_code, course_title, capacity, available_seats
    -> FROM COURSE
    -> WHERE course_id IN (
    ->     SELECT DISTINCT course_id
    ->     FROM COURSE_WAITLIST
    ->     WHERE status = 'ACTIVE')
    -> ;
+-------------+--------------------------------+----------+-----------------+
| course_code | course_title                   | capacity | available_seats |
+-------------+--------------------------------+----------+-----------------+
| CSE2001     | Data Structures and Algorithms |       60 |               0 |
+-------------+--------------------------------+----------+-----------------+
1 row in set (0.00 sec)

mysql>
```